



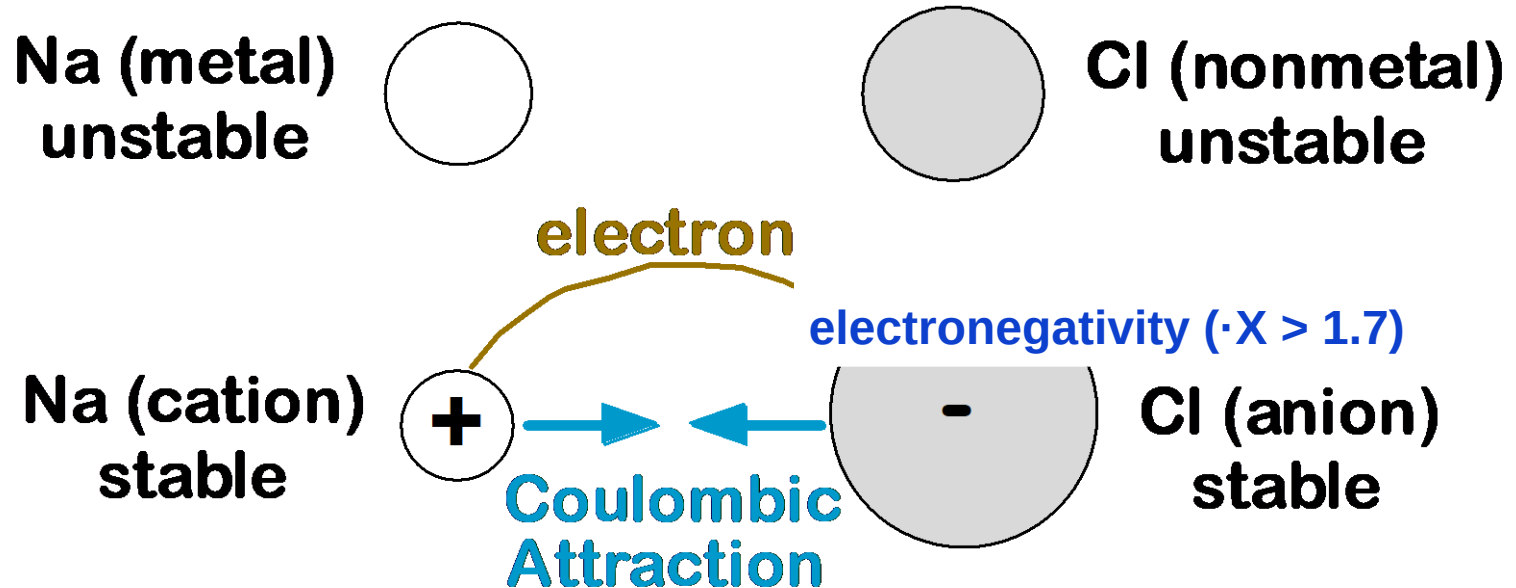
Outline:

- Types of bonds:
 - Ionic
 - Covalent
 - Metallic
 - Secondary bonding
- Examples:
 - *relation between bond energy and properties*
- Summary



IONIC BONDING

- Example: *NaCl*

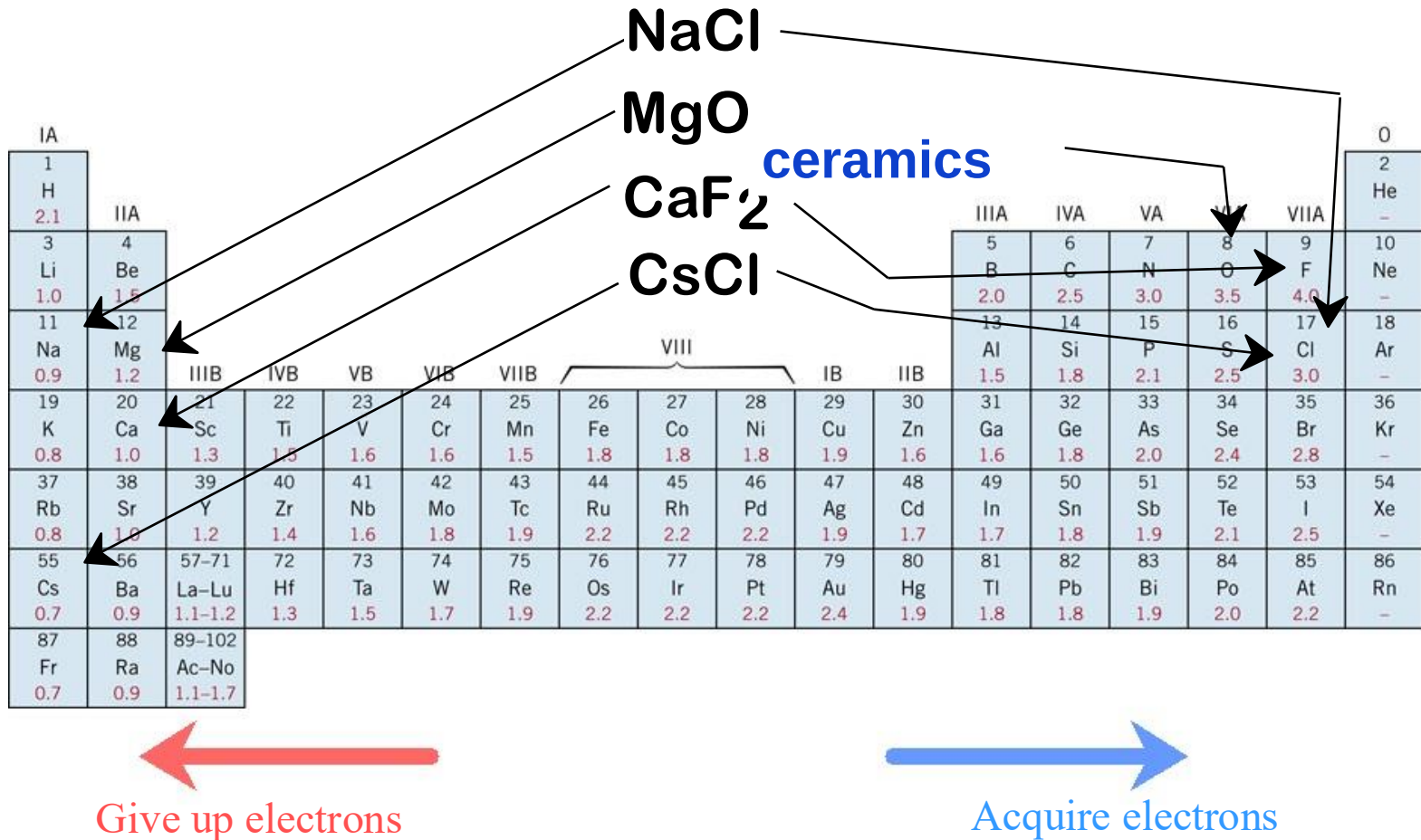


- Occurs between + and - ions.
- Requires electron transfer.
- Large difference in is required.



EXAMPLES: IONIC BONDING

- Predominant bonding in





IONIC BONDING: Summary

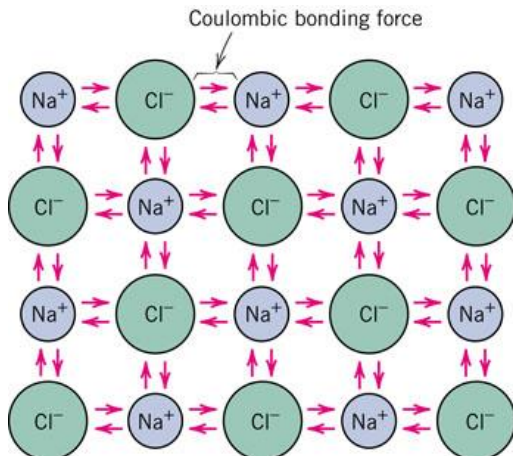
Ionic: electron transfer from one atom (cation) to the other (anion).

- More likely between atoms with **large electronegativity differences**

High ~~energy~~ found between metal and non-metal atoms:

NaCl, KF, CsBr, MgO...

Insulators / Low ~~energy~~ *ergies: 600 to 1500 kJ/mole (3 to 8 eV/atom)*



Typical characteristics of ionically-bonded materials:

- melting temperature
- Hard
- Brittle
- (electrical and thermal)



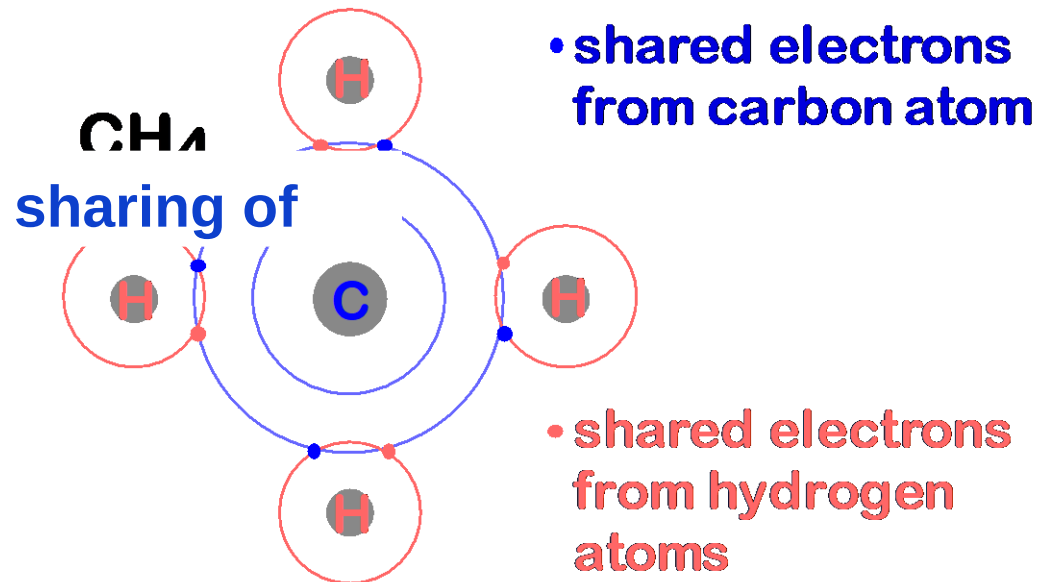
COVALENT BONDING

- Requires electrons
- Example: CH_4

C: has 4 valence e,
needs 4 more

H: has 1 valence e,
needs 1 more

*Electronegativities
are comparable.*



Adapted from Fig. 2.10, Callister 6e.



EXAMPLES: COVALENT BONDING

Diagram illustrating covalent bonding examples using the Periodic Table, showing electronegativity values and specific compounds highlighted.

Highlighted compounds and their electronegativity values:

- H_2 (Hydrogen, IA, 1.0)
- H_2O (Oxygen, VIA, 3.5)
- $C(\text{diamond})$ (Carbon, IVA, 2.5)
- SiC (Silicon Carbide, Si: 1.8, C: 2.5)
- F_2 (Fluorine, VIIA, 4.0)
- Cl_2 (Chlorine, VIIA, 3.0)
- $GaAs$ (Gallium Arsenide, Ga: 1.8, As: 2.0)

Periodic Table showing electronegativity values (Pauling scale) for elements:

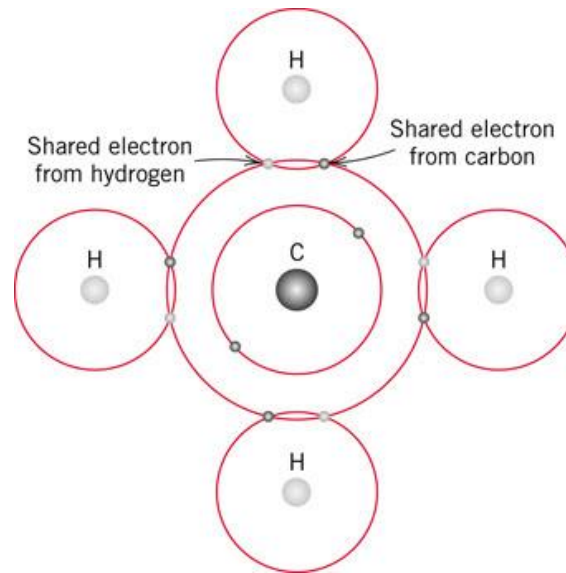
IA	IIA	IIIB	IVB	VB	VIB	VII B	VIII				IB	IIB	IIIA	column IVA	VA	VIA	VIIA	0
1 H 2.1	2 He -											3 Li 1.0	4 Be 1.5				10 Ne -	
11 Na 0.9	12 Mg 1.2											13 Al 1.5	14 Si 1.8	15 P 2.1	16 S 2.5	17 Cl 3.0	18 Ar -	
19 K 0.8	20 Ca 1.0	21 Sc 1.3	22 Ti 1.5	23 V 1.6	24 Cr 1.6	25 Mn 1.5	26 Fe 1.8	27 Co 1.8	28 Ni 1.8	29 Cu 1.9	30 Zn 1.6	31 Ga 1.6	32 Ge 1.8	33 As 2.0	34 Se 2.4	35 Br 2.8	36 Kr -	
37 Rb 0.8	38 Sr 1.0	39 Y 1.2	40 Zr 1.4	41 Nb 1.6	42 Mo 1.8	43 Tc 1.9	44 Ru 2.2	45 Rh 2.2	46 Pd 2.2	47 Ag 1.9	48 Cd 1.7	49 In 1.7	50 Sn 1.8	51 Sb 1.9	52 Te 2.1	53 I 2.5	54 Xe -	
55 Cs 0.7	56 Ba 0.9	57-71 La-Lu 1.1-1.2	72 Hf 1.3	73 Ta 1.5	74 W 1.7	75 Re 1.9	76 Os 2.2	77 Ir 2.2	78 Pt 2.2	79 Au 2.4	80 Hg 1.9	81 Tl 1.8	82 Pb 1.8	83 Bi 1.9	84 Po 2.0	85 At 2.2	86 Rn -	
87 Fr 0.7	88 Ra 0.9	89-102 Ac-No 1.1-1.7																

- Molecules with **nonmetals**
- Molecules with **metals** and **nonmetals**
- Elemental solids (**RHS** of Periodic Table)
- Compound solids (about **column IVA**)



COVALENT BONDING: *Summary*

Covalent: electron **sharing** between atoms each atom contributes (at least) one electron to the bond

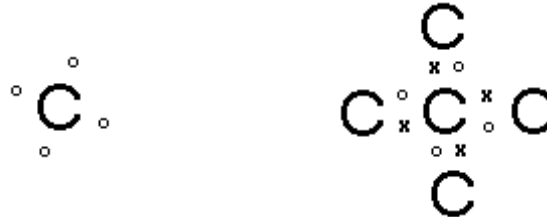


- Each atom tries to achieve a more stable orbital filling configuration
- Tends to be a highly **directional** bond
- Gives rise to a fixed orientation of the atoms
- Shared electrons may be considered to belong to each atom



COVALENT BONDING: *Summary*

Carbon:



Note how the sharing of electrons acts to complete the filling of electronic states in **each respective atom**:



Difficult to assign general characteristics to covalently-bonded materials:

- Bonds may be **strong** (diamond, $T_m > 3550^\circ\text{C}$) or **weak** (Bi, $T_m = 270^\circ\text{C}$)
- Materials may be *electrically* **conductive** (GaAs) or **insulating** (diamond)



Ionic Character

MOST MATERIALS ARE NEITHER 100% IONIC NOR 100% COVALENT

$$\% \text{ ionic character} = [1 - \exp\{-0.25(X_A - X_B)^2\}] \times 100\%$$

where X_A , X_B are the electronegativities of the A and B atoms, respectively.

Example:

Compute the percentage ionic character of the interatomic bonds for TiO_2 and ZnTe .

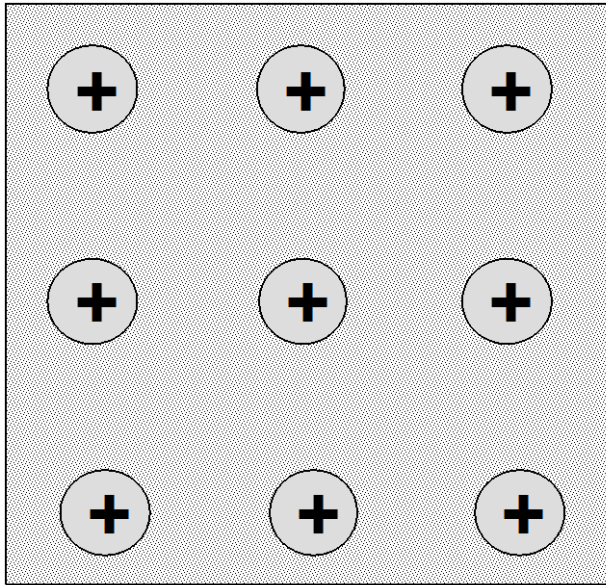
The electronegativities of the elements are found in Figure 2.7

- For TiO_2 , $X_{\text{Ti}} = 1.5$ and $X_{\text{O}} = 3.5$, and therefore,
- For ZnTe , $X_{\text{Zn}} = 1.6$ and $X_{\text{Te}} = 2.1$, and therefore,



METALLIC BONDING

- Arises from a sea of *donated* valence electrons (1, 2, or 3 from each atom).



Adapted from Fig. 2.11, *Callister 6e*.

- Valence electrons are not bound to any specific atom but are to drift throughout the material **free to drift**
- Active bonding electrons form an “*electron sea*”

- Primary bond for (*not surprisingly*) **metals** and their **alloys**



METALLIC BONDING

- Metallic bonding can be either **weak** (68 kJ/mole or 0.7 eV/atom for Hg) or **strong** (850 kJ/mole or 8.8 eV/atom for W)
- Metallic bonding gives rise to high electrical and thermal conductivity
- Metallic bonding also gives rise to **ductility** (*at least more than in most covalent and ionic solids*). Think about why this might be so?.

*The electrons are loosely held since each atom has several unoccupied valence orbitals; it is relatively easy for the electrons to move about. In this manner the electrons allow atoms to **slide** past each other.*

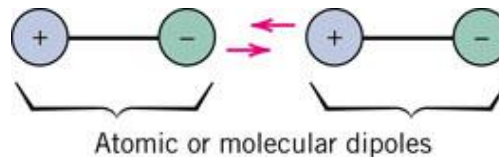


Secondary Bonding

- Van der Waals bonding
- Bond energy is very weak compared to others

Compare typical secondary bonding strengths (10 kJ/mole) with typical primary bonding strengths (50 to 1000 kJ/mole)

- Exists between almost all atoms and molecules
- Arise from atomic or molecular dipoles



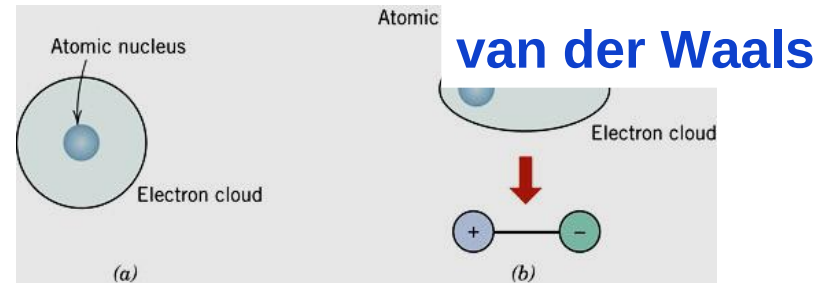
Physical bonds, not chemical



Secondary Bonding

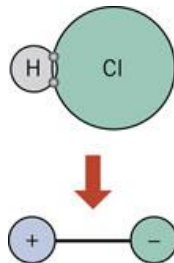
- Thermal vibration fluctuations can disrupt charge symmetry which leads to a dipole.

The presence of one dipole can induce a dipole in an adjacent molecule (or atom) and so on.



- Polar molecule induced dipoles & bonding

Due to permanent dipole moments in **hydrogen**



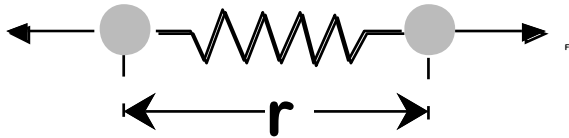
Unshielded, bare proton in H-O, H-F and H-N bonds lead to strong, hydrogen bonding

- **Hydrogen bonding** is a special case of secondary bonding.
- The **hydrogen bond** is generally stronger than

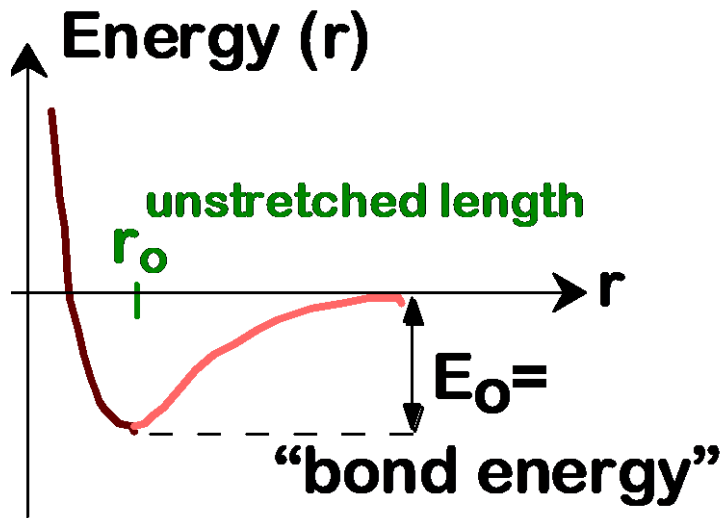


PROPERTIES FROM BONDING: T_m

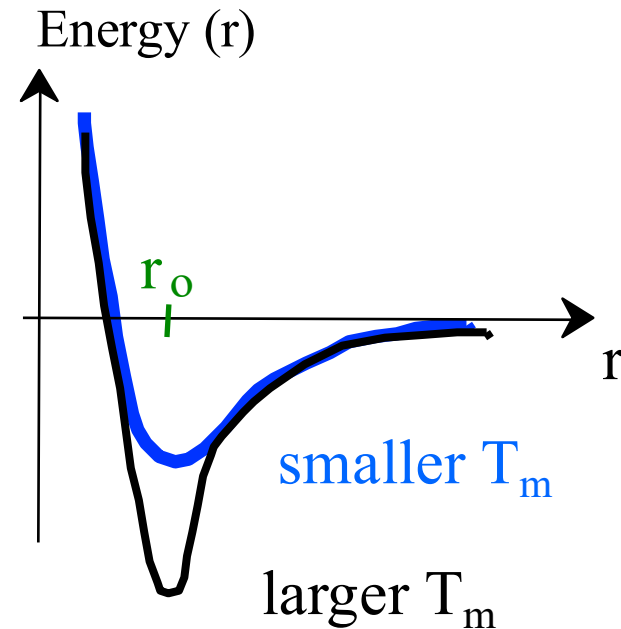
- Bond length, r



- Bond energy, E_o



- Melting Temperature, T_m

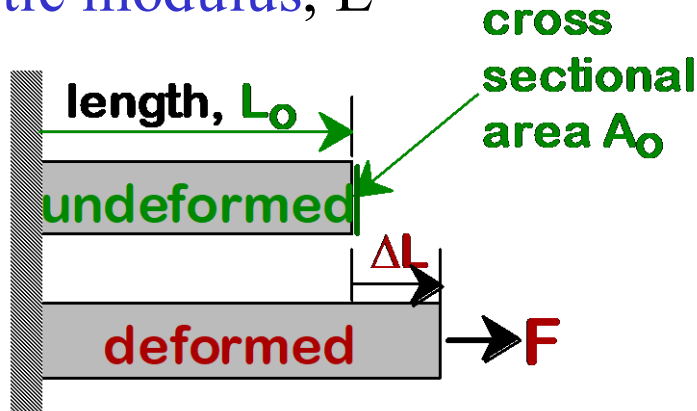


T_m is larger, if E_o is larger.



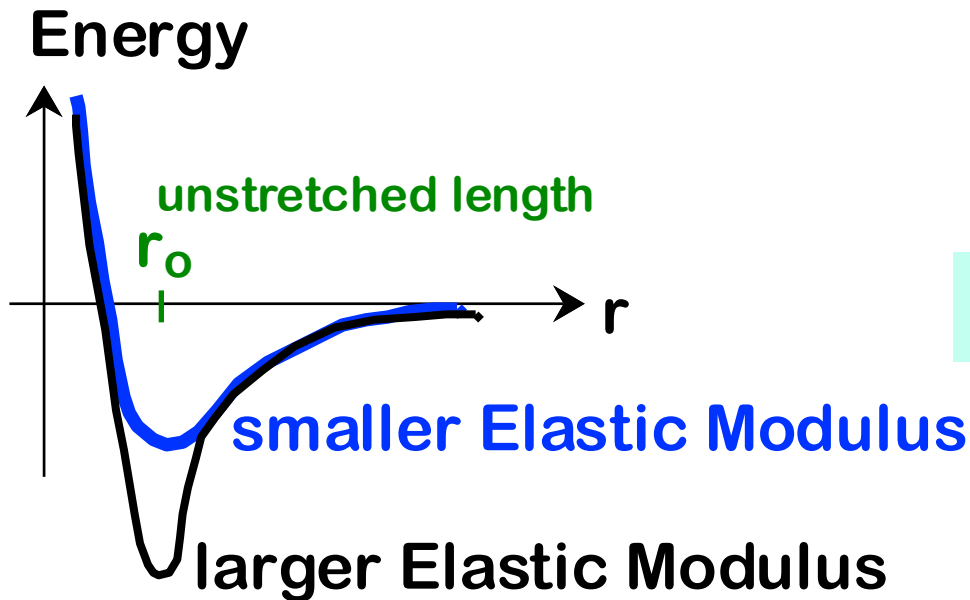
PROPERTIES FROM BONDING: E

- Elastic modulus, E



Elastic modulus

$$\frac{F}{A_0} = E \frac{\Delta L}{L_0}$$

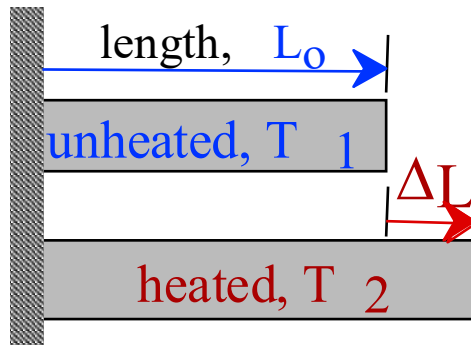


E is larger if E_0 is larger.



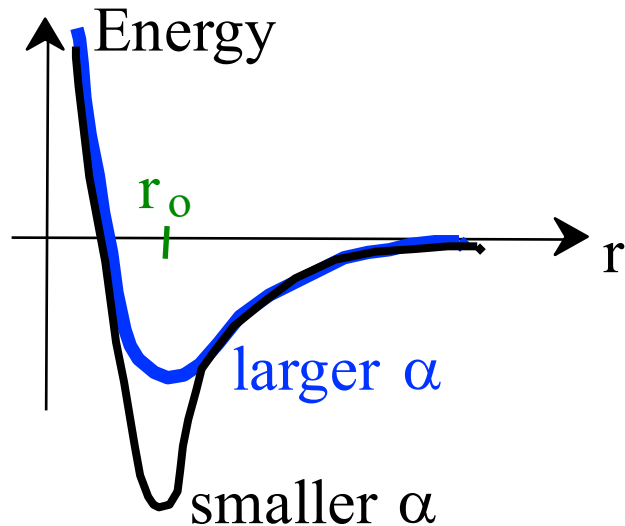
PROPERTIES FROM BONDING: α

- Coefficient of thermal expansion, α



coeff. thermal expansion

$$\frac{\Delta L}{L_0} = \alpha (T_2 - T_1)$$



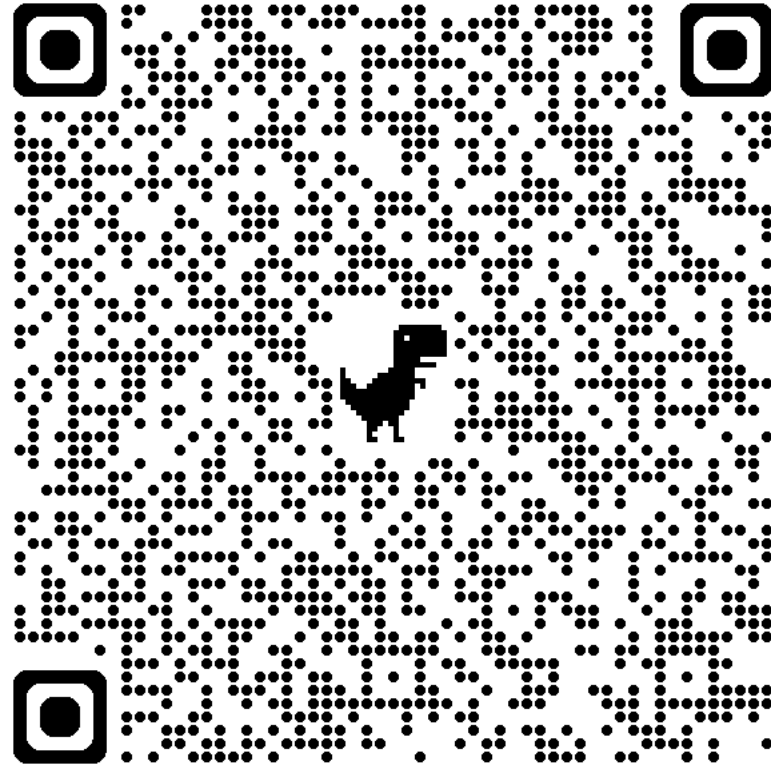
α is larger if E_0 is smaller.



Bonding Types: *Summary*

What is the bonding type in the following materials?

- Brass
- Rubber
- Solid Xenon
- AlP

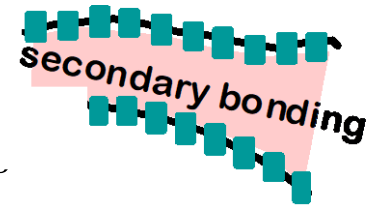




Bonding Types: Summary

A comparison of the type of bonding found in different materials:

- For brass, the bonding is since it is a metal alloy.
- For rubber, the bonding is with **some** **Metallic**.
Rubber is composed primarily of carbon and hydrogen
- For BaS, the bonding is predominantly **Covalent + van der Waals** character) - on the basis of the relative positions of Ba and S in the periodic table.
Ionic
- For solid xenon, the bonding is **van der Waals**
- For nylon, the bonding is with perhaps **some** **Covalent + Hydrogen**.
Nylon is composed primarily of carbon and hydrogen
- For AlP the bonding is predominantly (b **Covalent** character) on the basis of the relative positions of Al and P in the periodic table.





Next time:

Crystal structure, lattice directions and planes